

# A Complete Stationary Phase Atlas for the Noisy Mean-Field Kuramoto Equation

HCS-C347 source-local reconstruction

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## Abstract

For identical frequencies, positive noise, and attractive sinusoidal mean-field coupling, one theorem joins global classical probability flow, an exact free-energy dissipation law, exhaustion of stationary densities by von Mises profiles, the sharp synchronization threshold, the complete uniform Fourier spectrum, and a two-term critical expansion. The decisive Bessel quotient is proved strictly decreasing by an explicit positive-series coefficient pairing, which simultaneously closes the strict Turán inequality. Exact rational ledgers certify finite Bessel tails, stationary root brackets, and Fourier conventions; they do not replace the analytic continuum proof. No priority or arithmetic-target claim is made.

**Revision certificate.** Round-two critical expansion and evidence closure.

## 1 Frozen equation and main result

Let  $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ , use Lebesgue measure  $d\theta$ , and normalize  $\int_{\mathbb{T}} p d\theta = 1$ . Fix  $D > 0$ ,  $K \geq 0$ , and define

$$z[p] = \int_{\mathbb{T}} e^{i\theta} p(\theta) d\theta = r[p]e^{i\psi[p]}, \quad b[p](\theta) = Kr[p] \sin(\psi[p] - \theta).$$

The phase is immaterial when  $r = 0$ . Our equation is

$$\partial_t p = Dp_{\theta\theta} - \partial_{\theta}(b[p]p). \quad (1)$$

**Theorem 1** (Noisy identical-frequency atlas). *Let  $0 < \gamma < 1$  and let nonnegative  $p_0 \in C^{2+\gamma}(\mathbb{T})$  have unit mass.*

- (i) *Equation (1) has a unique global classical probability solution. It is strictly positive for  $t > 0$ .*
- (ii) *For positive times the free energy*

$$\mathcal{F}[p] = D \int_{\mathbb{T}} p \log p d\theta - \frac{K}{2} |z[p]|^2$$

*satisfies*

$$\mathcal{F}[p(t)] - \mathcal{F}[p(s)] = - \int_s^t \int_{\mathbb{T}} p |\partial_{\theta}(D \log p - KC[p])|^2 d\theta d\tau, \quad (2)$$

*where  $C[p](\theta) = \int \cos(\theta - \phi)p(\phi)d\phi$ . One may take  $s = 0$  if  $p_0 > 0$ .*

- (iii) *Every nonnegative  $C^2$  stationary probability density is uniform,  $p_* = 1/(2\pi)$ , or belongs to*

$$q_{\kappa, \psi}(\theta) = \frac{e^{\kappa \cos(\theta - \psi)}}{2\pi I_0(\kappa)}, \quad \kappa = \frac{K I_1(\kappa)}{D I_0(\kappa)}. \quad (3)$$

*For  $K \leq 2D$ , only  $p_*$  exists. For  $K > 2D$ , there is exactly one  $\kappa > 0$ , and  $\psi \in \mathbb{T}$  gives its complete rotation orbit.*

- (iv) *At  $p_*$ , the constant mass mode has eigenvalue zero, the real first-harmonic plane has eigenvalue  $K/2 - D$ , and the real harmonic plane of order  $n \geq 2$  has eigenvalue  $-Dn^2$ . The constant direction is removed on the probability tangent space.*

- (v) *If  $\delta = K/D - 2 \downarrow 0$  along the nonuniform branch, then*

$$\kappa^2 = 4\delta + \frac{2}{3}\delta^2 + O(\delta^3), \quad r^2 = \delta - \frac{5}{6}\delta^2 + O(\delta^3).$$

## 2 Global probability flow and dissipation

The map  $p \mapsto b[p]$  uses only two first Fourier coefficients and is smooth and locally Lipschitz between the relevant Hölder spaces. Standard heat-semigroup iteration gives a unique local classical solution. Mass conservation follows by integrating (1); comparison preserves nonnegativity, and the strong maximum principle gives strict positivity for positive time.

This local solution cannot blow up at finite time. For probability densities,  $\|\partial_\theta^j b[p]\|_\infty \leq K$  for every  $j$ . At a spatial maximum of

$$p_t = Dp_{\theta\theta} - b[p]p_\theta - (\partial_\theta b[p])p$$

we obtain  $\partial_t \max p \leq K \max p$ , whence  $\|p(t)\|_\infty \leq e^{Kt} \|p_0\|_\infty$  on finite intervals. Periodic parabolic Schauder estimates now give continuation through every finite time.

Because  $\partial_\theta C[p] = b[p]/K$ , with the zero-coupling case read directly, equation (1) is

$$p_t = \partial_\theta [p \partial_\theta (D \log p - KC[p])]. \quad (2)$$

The first variation of  $-K|z[p]|^2/2$  is  $-KC[p]$ . Multiplying (2) by the chemical potential and integrating by parts proves (2). Starting at  $s > 0$  handles initial zeros.

### 3 Stationary-flux exhaustion

A nonnegative stationary  $C^2$  density is positive by the elliptic strong maximum principle. Its flux

$$J = b[p]p - Dp_\theta$$

is constant. Division by  $p$  and one circuit around  $\mathbb{T}$  give

$$0 = \int \frac{p_\theta}{p} = \frac{1}{D} \int b[p] - \frac{J}{D} \int \frac{d\theta}{p(\theta)}.$$

The sine drift integrates to zero and the last integral is positive, so  $J = 0$ . Therefore

$$\partial_\theta \log p = \frac{Kr}{D} \sin(\psi - \theta) = \partial_\theta \left( \frac{Kr}{D} \cos(\theta - \psi) \right).$$

Normalization gives the von Mises form in (3). Conversely its first moment is

$$z[q_{\kappa, \psi}] = e^{i\psi} R(\kappa), \quad R(\kappa) = I_1(\kappa)/I_0(\kappa),$$

so zero flux is equivalent to  $\kappa = (K/D)R(\kappa)$ . This proves the stationary list in both directions.

### 4 Strict Turán inequality and the phase threshold

**Lemma 1.** *The function  $q(\kappa) = I_1(\kappa)/(\kappa I_0(\kappa))$  decreases strictly from  $1/2$  to zero on  $(0, \infty)$ . Equivalently,*

$$I_1(\kappa)^2 - I_0(\kappa)I_2(\kappa) > 0. \quad (3)$$

*Proof.* Put  $x = \kappa^2$ ,  $A(x) = I_1(\sqrt{x})/\sqrt{x}$ , and  $B(x) = I_0(\sqrt{x})$ . Their positive entire coefficients satisfy

$$A = \sum_{m \geq 0} a_m x^m, \quad B = \sum_{m \geq 0} b_m x^m, \quad \frac{a_m}{b_m} = \frac{1}{2(m+1)}.$$

Pairing indices in the derivative numerator yields

$$A'B - AB' = \sum_{m > n} (m-n)(a_m b_n - a_n b_m) x^{m+n-1} < 0. \quad (4)$$

Thus  $q = A/B$  is strictly decreasing and  $q(0+) = 1/2$ . Moreover  $0 < R(\kappa) < 1$ , since it is the strict expectation of  $\cos \theta$  under a positive von Mises density, so  $q(\kappa) < 1/\kappa \rightarrow 0$ .

Termwise Bessel recurrences give  $R' = 1 - R^2 - R/\kappa$  and  $I_2 = I_0 - 2I_1/\kappa$ . Consequently  $q' < 0$  is equivalent to  $\kappa(1 - R^2) < 2R$ , and multiplication by  $I_0^2/\kappa$  gives (3). Hence the strict Turán inequality is proved rather than assumed.  $\square$

For  $a = K/D > 0$ , a nonzero stationary solution obeys  $q(\kappa) = 1/a$ . Lemma 1 gives no solution for  $a \leq 2$  and exactly one for  $a > 2$ , completing the transition proof.

### 5 Complete uniform Fourier linearization

Write  $p = p_* + \varepsilon u$ ,  $\int u = 0$ . The linearization is

$$Lu = Du_{\theta\theta} - p_* \partial_\theta b[u], \quad b[u] = K \operatorname{Im}(z[u]e^{-i\theta}). \quad (5)$$

For  $u = a \cos \theta + c \sin \theta$ ,  $z[u] = \pi(a + ic)$ , and the last term in (5) is  $Ku/2$ . Every harmonic of order  $n \geq 2$  has zero first moment, leaving  $-Dn^2$ . Constants give mass conservation. This proves every eigenvalue and real multiplicity stated in Theorem 1.

## 6 Critical amplitude with a rigorous remainder

Exact series division gives

$$\frac{R(\kappa)}{\kappa} = \frac{1}{2} - \frac{\kappa^2}{16} + \frac{\kappa^4}{96} - \frac{11\kappa^6}{6144} + O(\kappa^8). \quad (6)$$

With  $x = \kappa^2$ , the left side of  $(2 + \delta)R(\sqrt{x})/\sqrt{x} = 1$  is analytic near the origin, and its  $x$ -derivative at  $(x, \delta) = (0, 0)$  is  $-1/8$ . The analytic implicit-function theorem applies. Coefficient matching gives

$$x = 4\delta + \frac{2}{3}\delta^2 + \frac{1}{18}\delta^3 + O(\delta^4).$$

Since  $\kappa = (2 + \delta)r$ ,

$$r^2 = \delta - \frac{5}{6}\delta^2 + \frac{43}{72}\delta^3 + O(\delta^4),$$

which proves the asserted expansion.

## 7 Exact evidence and boundaries

The deterministic receipt uses rational arithmetic only.

| finite panel                                     | rows |
|--------------------------------------------------|------|
| positive $I_0$ and $I_1/\kappa$ coefficients     | 17   |
| formal coefficients of $R/\kappa$                | 9    |
| certified positive-series Bessel tail brackets   | 7    |
| certified nonzero self-consistency root brackets | 4    |
| uniform real Fourier blocks                      | 162  |

After term 20, the positive-series tails are bounded by next-term geometric majorants. An independent checker reconstructs all 199 rows without importing the producer; a 60-check symbolic lane, two-directory byte replay, 71 hostile mutations, and strict JSON/YAML gates protect the artifact. The finite receipt does not prove the continuum theorem.

At  $K = 0$  the model is the heat equation. At  $K = 2D$  the first harmonic is neutral but Lemma 1 excludes nonuniform stationary profiles. Above threshold there is one concentration and an  $S^1$  phase orbit. The face  $D = 0$  is excluded: no atomic-state theorem is inferred. We claim neither general-initial-data convergence nor a rate, Hopf or time-periodic branches, disorder, delay, inertia, or finite-particle results.

## 8 Source and Route-A boundary

Sakaguchi’s noisy globally coupled oscillator paper[1] and the reversible mean-field rotator analysis of Bertini, Giacomini, and Pakdaman[2] fix the source lineage. The present derivation is a source-local reconstruction and makes no priority claim. The model has no arithmetic origin or prime clock, primitive arithmetic orbit ledger, target determinant, target analytic continuation, or natural target-zero quantization. Its tuple is

$$(A0_{\text{FAIL}}, A1_{\text{FAIL}}, A2_{\text{FAIL}}, A3_{\text{FAIL}}, A4_{\text{FAIL}}),$$

the verdict is Route-A rejected, and Route B is false. Bessel and Fourier data are not target arithmetic local data, Euler factors, root numbers, automorphy, a target divisor, target zeros, or a Hilbert–Pólya operator.

## References

- [1] H. Sakaguchi, Cooperative phenomena in coupled oscillator systems under external fields, *Progress of Theoretical Physics* **79** (1988), 39–46. <https://doi.org/10.1143/PTP.79.39>.
- [2] L. Bertini, G. Giacomini, and K. Pakdaman, Dynamical aspects of mean field plane rotators and the Kuramoto model, *Journal of Statistical Physics* **138** (2010), 270–290. <https://doi.org/10.1007/s10955-009-9908-9>.